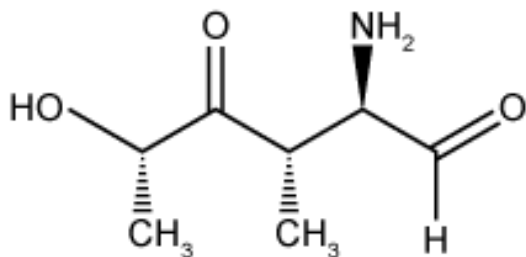


What is the correct systematic name for the compound shown below?



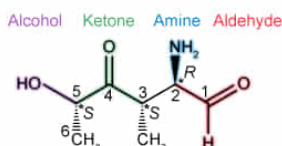
- ✓ ☒ A. (2*R*, 3*S*, 5*S*)-2-amino-5-hydroxy-3-methyl-4-oxohexanal [56%]
☐ B. (2*S*, 4*S*, 5*R*)-5-amino-2-hydroxy-4-methyl-6-oxohexan-3-one [11%]
☐ C. (1*S*, 3*S*, 4*R*)-4-amino-1,3-dimethyl-2,5-dioxopentan-1-ol [16%]
☐ D. (2*R*, 3*S*, 5*S*)-5-hydroxy-3,5-dimethyl-1,4-dioxopentan-2-amine [15%]

Correct

55% answered correctly

Explanation:

(2*R*, 3*S*, 5*S*)-2-amino-5-hydroxy-3-methyl-4-oxohexanal



	Increasing priority →			
Functional group				
Prefix	amino-	hydroxy-	oxo-	formyl
Suffix	-amine	-ol	-one	-al -carbaldehyde

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For compounds with multiple **functional groups**, the **highest-priority** functional group should be used as the basis for assigning a compound name. This **group** determines the compound **name ending (suffix)**, and all **lower-priority** functional groups are **named as substituent** branches using unique prefixes associated with each. Carbon atoms of the **longest continuous carbon chain** within the structure should be numbered to specify the location of the substituents. The selected carbon chain must incorporate the highest-priority functional group and must be numbered in the direction that assigns it the lowest possible number. **Stereocenter configurations** are also specified in parentheses using the same numbers at the beginning of the compound

name.

In the given compound, four functional groups are present, with an aldehyde group as the highest-priority group. Therefore, the compound is named as an aldehyde using the *-al* suffix. The longest continuous carbon chain containing the aldehyde is six carbon atoms long, and the chain is numbered to give the lowest position to the aldehyde carbon. The amine, ketone, and alcohol groups are subsequently located along the position reference numbers and are named as substituents using their respective prefixes. The configurations of the three stereogenic carbon atoms are then specified by position number at the beginning of the name to give (2*R*, 3*S*, 5*S*)-2-amino-5-hydroxy-3-methyl-4-oxohexanal.

(Choice B) The suffix and numbering for (2*S*, 4*S*, 5*R*)-5-amino-2-hydroxy-4-methyl-6-oxohexan-3-one incorrectly treat the ketone as the highest-priority group. Also, if an aldehyde substituent were present, the correct prefix should be *formyl*, not *oxo*.

(Choice C) The suffix and numbering for (1*S*, 3*S*, 4*R*)-4-amino-1,3-dimethyl-2,5-dioxopentane-1-ol incorrectly treat the *alcohol* as the highest-priority group, and five carbon atoms is not the longest chain containing the highest ranked group.

(Choice D) The suffix and numbering for (2*R*, 3*S*, 5*S*)-5-hydroxy-3,5-dimethyl-1,4-dioxopentane-2-amine incorrectly treat the *amine* as the highest-priority group. The 5-methyl is *not* a substituent because it is part of the longest continuous carbon chain (ie, the longest chain is six carbons rather than five carbons).

Educational objective:

Organic compounds with multiple functional groups are named using the highest-priority functional group to determine the suffix. All other functional groups are named as substituents using prefixes. Numbering of the longest carbon chain provides location references for substituents and stereocenter configurations and must assign the lowest possible number to the highest-priority group.

Time spent: 0 sec
Subject:
Organic Chemistry

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System:
Introduction to Organic Chemistry